

Global Superstore Sales Analysis

1. Project Overview

This project analyses the Global Superstore sales dataset containing orders from multiple countries and markets. The objective is to identify sales trends, customer purchasing behaviour, product performance, and profitability using SQL, Python, and Power BI. The insights generated help support data-driven business decisions and improve overall sales performance.

2. Dataset Summary

- **Rows:** 51,290
- **Columns:** 27 (before cleaning and feature engineering)
- **Key Features:** Customer details (Customer ID, Customer Name, Segment), Product details (Category, Sub-Category, Product Name), Sales metrics (Sales, Profit, Discount, Quantity, Shipping Cost), and Location details (City, State, Country, Region, Market).
- **Missing Data:** No missing values after data cleaning.

3. Exploratory Data Analysis using Python

Exploratory Data Analysis (EDA) was performed using Python to understand the structure and characteristics of the Global Superstore dataset.

- **Data loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `df.describe()` for summary statistics.

	discount	记录数	profit	quantity	row_id	sales	shipping_cost	year	weeknum
count	51290.000000	51290.0	51290.000000	51290.000000	51290.000000	51290.000000	51290.000000	51290.000000	51290.000000
mean	0.142908	1.0	28.610982	3.476545	25645.50000	246.498440	26.375818	2012.777208	31.287112
std	0.212280	0.0	174.340972	2.278766	14806.29199	487.567175	57.296810	1.098931	14.429795
min	0.000000	1.0	-6599.978000	1.000000	1.00000	0.000000	0.002000	2011.000000	1.000000
25%	0.000000	1.0	0.000000	2.000000	12823.25000	31.000000	2.610000	2012.000000	20.000000
50%	0.000000	1.0	9.240000	3.000000	25645.50000	85.000000	7.790000	2013.000000	33.000000
75%	0.200000	1.0	36.810000	5.000000	38467.75000	251.000000	24.450000	2014.000000	44.000000
max	0.850000	1.0	8399.976000	14.000000	51290.00000	22638.000000	933.570000	2014.000000	53.000000

```

<class 'pandas.DataFrame'>
RangeIndex: 51290 entries, 0 to 51289
Data columns (total 27 columns):
#   Column                Non-Null Count  Dtype
---  -
0   category              51290 non-null  str
1   city                  51290 non-null  str
2   country               51290 non-null  str
3   customer_id           51290 non-null  str
4   customer_name         51290 non-null  str
5   discount              51290 non-null  float64
6   market                51290 non-null  str
7   记录数                51290 non-null  int64
8   order_date            51290 non-null  str
9   order_id              51290 non-null  str
10  order_priority        51290 non-null  str
11  product_id            51290 non-null  str
12  product_name          51290 non-null  str
13  profit                51290 non-null  float64
14  quantity              51290 non-null  int64
15  region                51290 non-null  str
16  row_id                51290 non-null  int64
17  sales                 51290 non-null  int64
18  segment               51290 non-null  str
19  ship_date             51290 non-null  str
20  ship_mode             51290 non-null  str
21  shipping_cost         51290 non-null  float64
22  state                 51290 non-null  str
23  sub_category          51290 non-null  str
24  year                  51290 non-null  int64
25  market2              51290 non-null  str
26  weeknum               51290 non-null  int64
dtypes: float64(3), int64(6), str(18)
memory usage: 10.6 MB

```

- **Missing Data Handling:** Checked for null values and there are no null values
- **Duplicate Records:** Checked for duplicate records and there are no duplicate records.
- **Column Standardization:** Checked column names and renamed them to snake case for better readability and documentation.

```

Index(['category', 'city', 'country', 'customer_id', 'customer_name',
      'discount', 'market', '记录数', 'order_date', 'order_id', 'order_priority',
      'product_id', 'product_name', 'profit', 'quantity', 'region', 'row_id',
      'sales', 'segment', 'ship_date', 'ship_mode', 'shipping_cost', 'state',
      'sub_category', 'year', 'market2', 'weeknum'],
      dtype='str')

```

- **Feature Engineering:** Created new features **Shipping Days** (calculated as the difference between Ship Date and Order Date).

- **Data Consistency Check:** Removed the unnecessary **Unnamed 26** column, verified data types, checked for duplicate records, and confirmed there were no missing values.
- **Database Integration:** Loaded the cleaned dataset into **MySQL** for SQL queries and further business analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

- **Countries generating highest sales but low profit** - Identifies countries where strong sales are not translating into profitability, highlighting areas for cost optimization.

country	total_sales	total_profit
Turkey	108524	-98447.23
Philippines	183433	-16128.22
Dominican Republic	126137	-7613.5
Indonesia	404887	15608.68
New Zealand	172022	16600.28
Italy	289717	19828.76
Japan	100789	24328.47
Iran	113750	26856.24
Guatemala	131609	27944.69
Brazil	361098	30090.5
Nicaragua	149694	33401.44
Cuba	158861	38889.22
El Salvador	177565	42023.24
Spain	287162	54390.12
Mexico	622620	102818.1
Australia	925257	103907.43
Germany	628857	107322.82
France	858930	109029
United Kingdom	528570	111900.15
India	589664	129071.83
China	700591	150683.09
United States	2297354	286397.02

- **Top 10 most profitable products** - Determines the products that contribute the most to overall profit and revenue.

product_name	total_revenue	total_profit
Canon imageCLASS 2200 Advanced Copier	61600	25199.93
Cisco Smart Phone, Full Size	76441	17238.52
Motorola Smart Phone, Full Size	73159	17027.11
Hoover Stove, Red	31664	11807.97
Sauder Classic Bookcase, Traditional	39110	10672.07
Harbour Creations Executive Leather Armchair, ...	50120	10427.33
Nokia Smart Phone, Full Size	71904	9938.2
Cisco Smart Phone, with Caller ID	43124	9786.64
Nokia Smart Phone, with Caller ID	47880	9465.33
Belkin Router, USB	23473	8955.02

- **Loss-making sub-categories** - Identifies product sub-categories that consistently generate negative profits.

sub_category	total_revenue	total_profit
Tables	757034	-64083.39

- **Category with highest profit margin** - Compares product categories to find which delivers the highest profit relative to sales.

category	total_revenue	total_profit	profit_margin_percent
Technology	4744691	663778.73	13.99
Office Supplies	3787330	518473.83	13.69
Furniture	4110884	285204.72	6.94

- **Most profitable shipping mode** - Evaluates which shipping method contributes the highest overall profit.

ship_mode	total_profit
Standard Class	890596.02
Second Class	292583.53
First Class	208104.68
Same Day	76173.07

- **Top 5 highest-spending customers in each market** - Identifies the most valuable customers across different markets based on total spending.

market	customer_id	customer_name	total_spend
Africa	BW-10651	Barry Weirich	8957
Africa	LT-71101	Liz Thompson	7853
Africa	PO-91951	Phillina Ober	7799
Africa	JF-54151	Jennifer Ferguson	6470
Africa	JW-52201	Jane Waco	6173
APAC	CS-118451	Cari Sayre	16655
APAC	VG-218051	Vivek Grady	15654
APAC	PF-191201	Peter Fuller	15063
APAC	BF-110051	Barry Franz	14563
APAC	EM-139601	Eric Murdock	14324
Canada	SV-108152	Stuart Van	4009
Canada	DW-34802	Dianna Wilson	2469
Canada	JK-56252	Jim Karlsson	2264
Canada	BF-10802	Bart Folk	2183
Canada	DC-28502	Dan Campbell	2091
EMEA	SH-99752	Sally Hughsby	7537
EMEA	MD-78601	Michael Dominguez	6855
EMEA	PJ-90152	Pauline Johnson	5990
EMEA	CC-23702	Christopher Conant	5544
EMEA	TB-112502	Tim Brockman	5314
EU	SP-209202	Susan Pistek	16566
EU	DL-133152	Delfina Latchford	13191
EU	PO-188652	Patrick O'Donnell	13056
EU	ZC-219102	Zuschuss Carroll	11854
EU	JH-158202	John Huston	11729
LATAM	FH-143653	Fred Hopkins	13328
LATAM	MC-176353	Matthew Clasen	10126
LATAM	MD-173503	Maribeth Dona	9657
LATAM	RB-197953	Ross Baird	9463
LATAM	DV-130453	Darrin Van Huff	9018
US	SM-203204	Sean Miller	25042
US	TC-209804	Tamara Chand	19050
US	RB-193604	Raymond Buch	15117
US	TA-213854	Tom Ashbrook	14596
US	AB-101054	Adrian Barton	14476

- **Average shipping duration by ship mode** - Measures delivery performance by calculating average shipping time for each shipping method.

ship_mode	total_orders	average_shipping_duration
Standard Class	30775	5.00
Second Class	10309	3.23
First Class	7505	2.18
Same Day	2701	0.04

- **Average shipping time by market** - Compares delivery efficiency across different geographical markets.

market	total_orders	average_shipping_time
EU	10000	4.01
LATAM	10294	4.00
APAC	11002	3.97
US	9994	3.96
EMEA	5029	3.93
Africa	4587	3.91
Canada	384	3.68

- **Best-selling product in each sub-category** - Identifies the highest revenue-generating product within every product sub-category.

product_name	sub_category	total_sales
Belkin Router, USB	Accessories	23473
Hoover Stove, White	Appliances	32845
Boston Canvas, Fluorescent	Art	7804
Fellowes PB500 Electric Punch Plastic Comb Bind...	Binders	27454
Sauder Classic Bookcase, Traditional	Bookcases	39110
Hon Executive Leather Armchair, Adjustable	Chairs	58200
Canon imageCLASS 2200 Advanced Copier	Copiers	61600
Ames Interoffice Envelope, Security-Tint	Envelopes	2840
Stockwell Paper Clips, Assorted Sizes	Fasteners	2404
Rubbermaid Frame, Duo Pack	Furnishings	9838
Dot Matrix Printer Tape Reel Labels, White, 500...	Labels	3324
Okidata Inkjet, Wireless	Machines	23003
Green Bar Cards & Envelopes, Multicolor	Paper	4308
Apple Smart Phone, Full Size	Phones	86936
Eldon File Cart, Single Width	Storage	34392
High Speed Automatic Electric Letter Opener	Supplies	17031
Hon Conference Table, Adjustable Height	Tables	15018

- **Monthly sales and profit (MoM growth)** - Analyses monthly revenue, profit, and growth trends to monitor business performance.

year	month	total_revenue	total_profit	profit_margin	revenue_mom_growth_percent	profit_mom_growth_percent
2014	1	241267	28001.39	11.61	NULL	NULL
2014	2	184848	19751.7	10.69	-23.38	-29.46
2014	3	263110	37357.26	14.2	42.34	89.13
2014	4	242781	23782.3	9.8	-7.73	-36.34
2014	5	288404	33953.56	11.77	18.79	42.77
2014	6	401843	43778.6	10.89	39.33	28.94
2014	7	258718	28035.87	10.84	-35.62	-35.96
2014	8	456633	53542.89	11.73	76.50	90.98
2014	9	481186	67979.45	14.13	5.38	26.96
2014	10	422785	58209.83	13.77	-12.14	-14.37
2014	11	555312	62856.59	11.32	31.35	7.98
2014	12	503154	46916.52	9.32	-9.39	-25.36

- **Year-over-year sales growth** - Evaluates annual business growth by comparing revenue and profit across years.

year	total_revenue	total_profit	profit_margin	revenue_yoy_growth_percent	profit_yoy_growth_percent
2011	2259511	248940.81	11.02	NULL	NULL
2012	2677493	307415.28	11.48	18.50	23.49
2013	3405860	406935.23	11.95	27.20	32.37
2014	4300041	504165.97	11.72	26.25	23.89

- **Products with highest average discount** - Identifies products receiving the largest average discounts to assess pricing strategies.

product_name	average_discount
Chromcraft Training Table, Adjustable Height	0.8
Eureka Disposable Bags for Sanitaire Vibra Groo...	0.8
GBC Plasticlear Binding Covers	0.73
GBC VeloBinder Electric Binding Machine	0.73
Zebra GK420t Direct Thermal/Thermal Transfer ...	0.7
Brother MFC-9340CDW LED All-In-One Printer, ...	0.7
Epson Perfection V600 Photo Scanner	0.7
Bush Westfield Collection Bookcases, Dark Cher...	0.7
Plantronics Single Ear Headset	0.7
Okidata B401 Printer	0.7

- **Top loyal (repeat) customers in each market** - Identifies customers with the highest repeat purchase frequency and spending.

market	customer_id	customer_name	total_orders	total_spend
Africa	AS-6301	Ann Steele	8	1908
Africa	MA-79951	Michelle Arnett	8	1558
Africa	AA-6451	Anna Andreadi	8	1503
APAC	DK-130901	Dave Kipp	18	10186
APAC	CS-121751	Charles Sheldon	17	13516
APAC	EM-139601	Eric Murdock	16	14324
Canada	CC-22202	Chris Cortes	5	944
Canada	DW-34802	Dianna Wilson	2	2469
Canada	DC-28502	Dan Campbell	2	2091
EMEA	TB-112502	Tim Brockman	10	5314
EMEA	DV-30452	Darrin Van Huff	10	3374
EMEA	DK-31502	David Kendrick	10	1246
EU	JG-158052	John Grady	15	7843
EU	AR-108252	Anthony Rawles	15	7720
EU	YC-218952	Yoseph Carroll	14	8531
LATAM	TZ-214453	Tom Zandusky	15	7853
LATAM	JH-159853	Joseph Holt	15	3916
LATAM	RB-197953	Ross Baird	14	9463
US	EP-139154	Emily Phan	17	5481
US	ZC-219104	Zuschuss Carroll	13	8028
US	JE-157454	Joel Eaton	13	6762

- **Highest revenue city in each country** - Finds the top-performing city in each country based on total sales revenue.

country	city	total_revenue
United States	New York City	256397
Philippines	Manila	120896
Australia	Sydney	101952
Indonesia	Jakarta	94318
Mexico	Mexico City	85732
Nicaragua	Managua	83714
United Kingdom	London	80766
Dominican Republic	Santo Domingo	78710
Thailand	Bangkok	76045
France	Paris	63316
Austria	Vienna	62026
Germany	Berlin	61918
El Salvador	San Salvador	57708
Spain	Madrid	51387
Honduras	Tegucigalpa	48061
Guatemala	Mixco	47901
Vietnam	Ho Chi Minh City	43263
Democratic Repub...	Kinshasa	42533
Bangladesh	Dhaka	41424
Singapore	Singapore	40285
Brazil	São Paulo	38972
Egypt	Cairo	38436
Myanmar (Burma)	Yangon	34144
Pakistan	Lahore	33775
Saudi Arabia	Riyadh	33031
Italy	Rome	32107
China	Beijing	31637
India	Delhi	31014
Morocco	Casablanca	29798
Iraq	Baghdad	28875
Argentina	Buenos Aires	28577
Cuba	Santiago de C...	24532
South Africa	Johannesburg	23256
Senegal	Dakar	22274
Turkey	Istanbul	21896
Panama	Panama City	21534
Colombia	Bogotá	21112
New Zealand	Auckland	20154
Sweden	Stockholm	17498
Cambodia	Phnom Penh	17476
Nigeria	Lagos	17188
Malaysia	Kuala Lumpur	17176
Afghanistan	Kabul	16951
Japan	Tokyo	16219
Cote d'Ivoire	Abidjan	16189
Chile	Santiago	15823
Ghana	Accra	15555
Angola	Luanda	14679
South Korea	Seoul	14304

Iran	Mashhad	13470
Zambia	Lusaka	12764
Cameroon	Yaounde	12154
Tanzania	Dar es Salaam	11843
Belgium	Ghent	11737
Ukraine	Kharkiv	11567
Russia	Chelyabinsk	11347
Ireland	Dublin	11071
Peru	Lima	11009
Switzerland	Zurich	10686
Haiti	Carrefour	9609
Madagascar	Antananarivo	9604
Algeria	Algiers	9268
Sudan	Khartoum	9204
Netherlands	Amsterdam	9122
Finland	Helsinki	8306
Canada	Montréal	8289
Mozambique	Maputo	8221
Bulgaria	Sofia	8108
Norway	Oslo	7805
Taiwan	Taipei	7648
Hungary	Budapest	7471
Kenya	Nairobi	7433
Barbados	Bridgetown	7175
Mongolia	Ulan Bator	6789
Mali	Bamako	6741
Lesotho	Maseru	6712
Ecuador	Quito	6557
Portugal	Ecuador	6217
Hong Kong	Kowloon	6148
Qatar	Doha	6050
Martinique	Fort-de-France	5968
Czech Republic	Prague	5943
Poland	Lublin	5934
Venezuela	Caracas	5648
Azerbaijan	Baku	5630
Uruguay	Montevideo	5607
Belarus	Hrodna	5115
Trinidad and Tobago	San Fernando	4916
Denmark	Copenhagen	4846
Niger	Niamey	4800
Rwanda	Kigali	4756
Jordan	Amman	4652
Moldova	Chisinau	4511
Somalia	Hargeysa	4330
Kyrgyzstan	Bishkek	4186
Estonia	Tallinn	4095
Mauritania	Nouakchott	4020

Montenegro	Podgorica	4004
Israel	Ashdod	3995
Libya	Misratah	3915
Syria	Homs	3635
Jamaica	Spanish Town	3532
Nepal	Kathmandu	3522
Liberia	Monrovia	3512
Romania	Galati	3377
Lithuania	Vilnius	3275
Georgia	Tbilisi	3265
Togo	Lome	3254
Croatia	Zagreb	3087
Papua New Guinea	Port Moresby	2928
Namibia	Windhoek	2899
Republic of the Co...	Loubomo	2714
Djibouti	Djibouti	2394
Central African Re...	Bangui	2376
Uzbekistan	Termez	2366
Bosnia and Herze...	Sarajevo	2273
Bolivia	Santa Cruz de...	2241
Benin	Porto-Novo	2130
Zimbabwe	Bulawayo	2083
Guinea-Bissau	Bissau	2065
Yemen	Taizz	2030
Tunisia	Tunis	1762
Gabon	Libreville	1752
Albania	Elbasan	1619
Lebanon	Beirut	1596
Slovenia	Ljubljana	1469
Guadeloupe	Les Abymes	1462
Chad	Koumra	1317
Sierra Leone	Koidu	1271
Guinea	Nzerekore	1223
Paraguay	San Lorenzo	1175
Uganda	Kampala	1152
Turkmenistan	Ashgabat	1138
South Sudan	Juba	1045
Kazakhstan	Pavlodar	853
Sri Lanka	Colombo	852
Ethiopia	Dire Dawa	850
Slovakia	Nitra	743
Bahrain	Al Muharraq	669
United Arab Emira...	Ras al Khaymah	533
Swaziland	Manzini	288
Burundi	Bujumbura	268
Tajikistan	Kulob	242
Macedonia	Kumanovo	199
Eritrea	Keren	188

- **Profit contribution by region** - Measures each region's percentage contribution to the company's total profit.

region	total_profit	profit_percent
Central	311403.98	21.22
North	194597.95	13.26
North Asia	165578.42	11.28
South	140355.77	9.56
Central Asia	132480.19	9.03
Oceania	120089.11	8.18
West	108418.45	7.39
East	91522.78	6.24
Africa	88871.63	6.06
EMEA	43897.97	2.99
Caribbean	34571.32	2.36
Southeast ...	17852.33	1.22
Canada	17817.39	1.21

- **Order priority analysis** - Evaluates sales, profit, and average order value across different order priority levels.

order_priority	orders	total_sales	total_profit	profit_percent	average_sales_per_order
Medium	29433	7281105	864203.76	58.89	247.38
High	15501	3807699	420373.51	28.65	245.64
Critical	3932	986258	124224.16	8.47	250.83
Low	2424	567843	58655.85	4	234.26

5. Dashboard in Power BI

Finally, an interactive dashboard was developed in Power BI to visualize key business insights. The dashboard includes KPIs, sales and profit trends, regional performance, customer analysis, product performance, and shipping insights. Interactive filters and slicers enable users to explore the data and support informed business decision-making.

Superstore Sales Dashboard

Year

☐ 2011

☐ 2012

☐ 2013

☐ 2014

Market

☐ Africa

☐ APAC

☐ EMEA

☐ EU

☐ LATAM

☐ North America

Category

☐ Furniture

☐ Office Supplies

☐ Technology

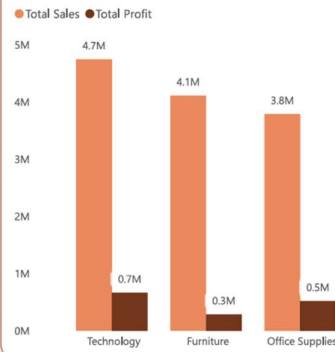
\$12.64M
Total Sales

\$1.47M
Total Profit

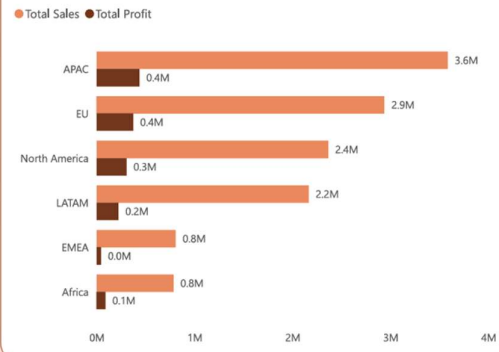
25,035
Total Orders

11.61%
Profit Margin %

Sales by category



Sales by market



Superstore Sales Dashboard

Year

☐ 2011

☐ 2012

☐ 2013

☐ 2014

Market

☐ Africa

☐ APAC

☐ EMEA

☐ EU

☐ LATAM

☐ North America

Category

☐ Furniture

☐ Office Supplies

☐ Technology

4,873

Total Customers

10,292

Total Products

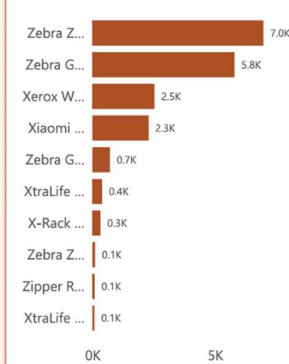
\$505.01

Average Order Value

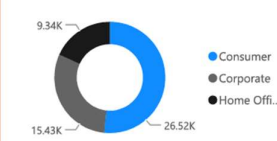
3.48

Average Quantity Sold

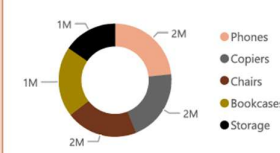
Top 10 Products by sales



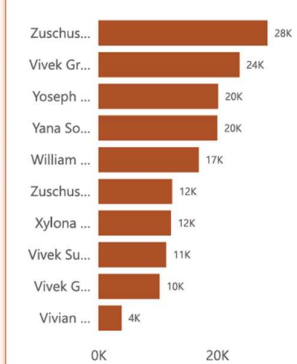
Customers by segment



Top 10 Sub-categories by sales



Top 10 Customers by sales





6. Key Insights

- **Technology** is the highest-performing product category, generating the **largest share of both sales and profit** compared to other categories.
- The APAC and EU markets generate **nearly equal profit**, despite APAC recording **approximately \$0.6 million higher sales**, indicating lower profit margins in APAC.
- **Higher discounts are associated with lower profits**, suggesting that excessive discounting negatively impacts profitability.
- Around **60% of customers choose Standard Class shipping**, even though it has an **average delivery time of more than 4 days**, indicating that customers prioritize lower shipping costs over faster delivery.

7. Business Recommendations

- **Improve Profitability:** Focus on countries and regions with high sales but low profit by optimizing pricing and operational costs.
- **Strengthen Customer Retention:** Reward high-value and repeat customers through loyalty programs and personalized offers.
- **Optimize Product Portfolio:** Promote top-performing products and review or discontinue consistently loss-making sub-categories.
- **Review Discount Strategy:** Reduce excessive discounts on products with low profit margins while maintaining competitive pricing.

- **Enhance Logistics:** Promote the most profitable shipping modes and improve delivery performance in markets with longer shipping times.

8. Conclusion

This project demonstrated how SQL, Python, and Power BI can be used together to analyse sales data and generate actionable business insights. The findings can help organizations improve profitability, optimize operations, and make informed business decisions.